NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS3 SUBJECT: Chemistry

TOPIC: Quantitative Chemistry & Stoichiometry

Introduction

- Quantitative chemistry deals with the amounts of substances in reactions: the mole, molar mass, molar volume, concentrations, and the calculations derived from balanced equations. It is 'the arithmetic of chemistry'.

Core relationships

- n = mass ÷ molar mass; n = volume (dm^3) ÷ 24 (at r.t.p.); concentration = n ÷ volume (dm^3); number of particles = n × 6.02×10^{23}. Stoichiometry: use the balanced equation's mole ratio. Percentage composition of a compound = mass of the element ÷ molar mass × 100. Empirical formula from the masses; molecular formula = n × empirical formula.

Worked example

- What mass of CaO is obtained from 50 g of $CaCO_3$ ($CaCO_3 \rightarrow CaO + CO_2$)? Molar masses: $CaCO_3$ = 100, CaO = 56. Moles $CaCO_3$ = 50/100 = 0.5; ratio 1:1 → 0.5 mol CaO; mass = 0.5 × 56 = 28 g.

Titration calculations

- In a titration, moles acid = moles alkali × (ratio). Example: 25.0 cm^3 of 0.100 mol/dm^3 NaOH neutralises 20.0 cm^3 of HCl; moles NaOH = 0.100 × 0.025 = 0.0025; moles HCl = 0.0025; concentration = 0.0025 ÷ 0.020 = 0.125 mol/dm^3.

CLASS EVALUATION

1. Calculate the number of moles in 22 g of CO_2 (C = 12, O = 16).
2. What mass of iron is in 40 g of Fe_2O_3 (Fe = 56, O = 16)?
3. Define the empirical formula.
4. 25.0 cm^3 of 0.08 mol/dm^3 H_2SO_4 neutralises 20.0 cm^3 of NaOH. Find the concentration of the NaOH ($H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$).

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WEEK: EIGHT

CLASS: SS3 SUBJECT: Chemistry

TOPIC: Practical Chemistry Skills (Volumetric & Descriptive)

Introduction

- Practical chemistry is examined both practically (the WASSCE practical paper) and descriptively. The skills: accurate weighing and measuring, titration, gas collection, the correct use of apparatus, safe heating, and clear recording of observations and inferences.

Apparatus and their use

- Burette (titrant delivery), pipette (fixed volumes, with a filler), conical flask (titrated solution), measuring cylinder (rough volumes), beaker, test tube and boiling tube, evaporating dish (crystallisation), crucible (strong heating), funnel, filter paper, watch glass, spatula, retort stand, Bunsen burner, water bath, gas jar. Read the meniscus at eye level; wash and rinse apparatus; repeat titrations to concordant values (within 0.1 cm^3).

Gas tests

- O_2 — rekindles a glowing splint; H_2 — a lighted splint gives a pop; CO_2 — turns limewater milky; NH_3 — turns damp red litmus blue and forms white fumes with HCl; SO_2 — turns acidified potassium dichromate green; Cl_2 — bleaches damp litmus; H_2S — blackens lead acetate paper.

Tests for ions

- Cations: flame tests (Na^+ — yellow; K^+ — lilac; Ca^{2+} — brick red; Cu^{2+} — blue-green), NaOH (Cu^{2+} — blue precipitate; Fe^{2+} — green then brown; Fe^{3+} — reddish-brown; Al^{3+}/Zn^{2+} — white, soluble in excess NaOH; NH_4^+ — ammonia on warming). Anions: Cl^- — white AgCl precipitate (soluble in ammonia (dilute)); SO_4^{2-} — white $BaSO_4$ (insoluble in HCl); CO_3^{2-} — effervescence with acid (CO_2); NO_3^- — brown ring.

CLASS EVALUATION

1. How do you test for hydrogen gas?
2. What is the flame colour of sodium? Of potassium?
3. Describe the test for a chloride ion.
4. Why must a pipette be rinsed with the solution it will measure?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS3 SUBJECT: Chemistry

TOPIC: Separation & Purification (Advanced)

Introduction

- Separation and purification methods exploit differences in the physical properties of the components of a mixture: particle size, solubility, boiling point, density and the adsorption of substances to a stationary phase.

Techniques and their theory

- **Filtration** — insolubility; **evaporation and crystallisation** — solubility differences (recrystallisation purifies a solid); **simple distillation** — different boiling points (water from salt); **fractional distillation** — close boiling points (air → N_2, O_2; crude oil → fractions; ethanol–water); **separating funnel** — immiscible liquids of different densities (oil and water); **chromatography (paper, thin-layer, column, gas)** — different affinities for the mobile and stationary phases (identify dyes, inks, food colouring; calculate Rf = distance moved by the component ÷ distance moved by the solvent); **centrifugation** — density (blood cells); **sublimation** — iodine, ammonium chloride; **magnetic separation and decantation**.

Purifying a product

- Choose the method by the properties; test the purity by melting/boiling point (a pure substance has a sharp, constant value) or by chromatography (a pure substance gives one spot).

CLASS EVALUATION

1. How would you separate a mixture of sand, salt and iodine?
2. What is Rf in chromatography?
3. Why does fractional distillation separate crude oil?
4. How is the purity of a solid checked?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS3 SUBJECT: Chemistry

TOPIC: Radiochemistry Basics

Introduction

- Radiochemistry is the chemistry of radioactive substances — elements whose nuclei are unstable and decay spontaneously, emitting radiation. The three types of radiation are alpha (α), beta (β) and gamma (γ).

Radiation and decay

- **Alpha** — a helium nucleus (2 protons + 2 neutrons): heavy, positive, weakly penetrating (stopped by paper). **Beta** — an electron from the nucleus: light, fast, penetrates a few mm of aluminium. **Gamma** — high-energy electromagnetic radiation: very penetrating (needs lead or concrete). Half-life ($t_{1/2}$) is the time for half the nuclei to decay: after one $t_{1/2}$ half remains, after two $t_{1/2}$ a quarter, after three an eighth.

Nuclear equations

- The mass and atomic numbers balance: ^{23}■■■U → ^{23}■■■Th + ■■He (alpha); ^{1}■■C → ^{1}■■N + ■■■e (beta). Isotopes decay at their own rates — C-14 dating uses the 5,730-year half-life.

Uses and safety

- Medicine: radiotherapy for cancer (Co-60, I-131 for the thyroid), diagnosis (tracers, imaging); industry: thickness gauges, sterilisation of equipment and food; agriculture: mutation breeding, pest control (sterile insect technique); archaeology: radiocarbon dating. Safety: time, distance and shielding; the ALARA principle; lead aprons, film badges and the careful disposal of sources.

CLASS EVALUATION

1. Name the three types of radiation and their penetrating powers.
2. Define the half-life.
3. Write the alpha decay equation of ^{22}■■■Ra.
4. State two uses of radioactivity in medicine.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS3 SUBJECT: Chemistry

TOPIC: Energy Changes (Thermochemistry) & Rates of Reaction

Introduction

- Chemical reactions involve energy and time. Thermochemistry studies the heat changes; kinetics studies the speed. Both are core WAEC/NECO topics — and both explain many everyday processes.

Exothermic and endothermic reactions

- **Exothermic** — heat is released (the products have less energy than the reactants; ΔH is negative): combustion, respiration, the reaction of sodium or calcium with water, neutralisation, the slaking of lime. **Endothermic** — heat is absorbed (ΔH is positive): photosynthesis, the decomposition of calcium carbonate and ammonium chloride, the dissolving of ammonium nitrate. Energy profiles show the activation energy — the minimum energy colliding particles need to react; a catalyst lowers it.

Measuring heat changes

- In the school laboratory: the simple calorimeter — a known mass of water (or solution) in a cup, the temperature measured before and after the reaction, and the heat calculated: $q = mc\Delta T$ (m = mass of water in g; c = 4.2 J g^{-1} K^{-1}; ΔT = the temperature change); then $\Delta H = -q \div$ moles of the limiting reactant (in kJ/mol).

Rates of reaction

- Rate = change in concentration/amount ÷ time — measured by the volume of gas collected, the loss of mass, or the time for a mark to disappear (the 'disappearing cross' with sodium thiosulphate and acid). Factors that increase the rate: **concentration** (more particles per volume — more frequent collisions), **temperature** (faster, more energetic collisions — about a doubling per 10 °C), **surface area** (crushing a solid), **pressure** (gases), and a **catalyst** (lowers the activation energy).

CLASS EVALUATION

1. Distinguish exothermic from endothermic reactions with two examples each.
2. What is activation energy?

3. Calculate the heat absorbed when 100 g of water warms from 25 to 40 °C.
4. State the four factors that affect the rate of a reaction.
5. Why does a powder react faster than a lump?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS3 SUBJECT: Chemistry

TOPIC: Acids, Bases & Salts (Revision & Applications)

Introduction

- Acids, bases and salts are the oldest families of chemistry — and the most examined. This week revises their definitions, properties, reactions and everyday uses at the WAEC/NECO level.

Acids

- Arrhenius: acids donate H■ in water; the Brønsted–Lowry definition: acids are proton donors. Properties: sour taste, turn blue litmus red, conduct electricity, react with metals (release H■), with carbonates (release CO■), with bases (neutralisation). Strong acids ionise fully (HCl, H■SO■, HNO■); weak acids partly (ethanoic, citric, carbonic). Uses: H■SO■ in car batteries and fertilisers, HCl in the stomach, HNO■ in explosives and fertilisers.

Bases and alkalis

- A base is a metal oxide or hydroxide that neutralises acids; an alkali is a base soluble in water (NaOH, KOH, Ca(OH)■, NH■ solution). Properties: soapy feel, turn red litmus blue, corrosive. Uses: slaked lime in building and agriculture (to neutralise acidic soil and for the treatment of sewage), caustic soda in soap making, ammonia in fertilisers and cleaning agents.

Salts and their preparation

- A salt is formed when the H■ of an acid is replaced by a metal or ammonium ion. Types: normal, acidic, basic, double and complex salts. Preparation: the acid + metal, + metal oxide, + carbonate (all by evaporation/crystallisation); acid + alkali (titration — to get a soluble salt with a known concentration); and double decomposition (for insoluble salts — e.g. AgCl, BaSO■, by mixing solutions and filtering). Solubility rules: all nitrates are soluble; chlorides are soluble except AgCl, PbCl■ and Hg■Cl■; sulphates soluble except BaSO■, PbSO■, CaSO■; carbonates are insoluble except those of sodium, potassium and ammonium.

CLASS EVALUATION

1. Define an acid and a base by the Brønsted–Lowry theory.

2. State three properties of acids.
3. Name the products of the reaction of an acid with a carbonate.
4. Which method prepares silver chloride?
5. Give one use each of H■SO■, NaOH and slaked lime.